

## TULLINGE, Sweden

WMO: 024690

Lat: 59.1785N

Lon: 17.9093E

Elev: 147

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |            | Humidification DP/MCDB and HR |            |             |            |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|------------|-------------------------------|------------|-------------|------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |            | 99.6%                         |            |             | 99%        |            |            | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%        | DP                            | HR         | MCDB        | DP         | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-2.2 | (c)<br>4.5 | (d)<br>-6.6                   | (e)<br>3.9 | (f)<br>-2.1 | (g)<br>0.7 | (h)<br>5.7 | (i)<br>4.9 | (j)<br>22.4           | (k)<br>33.5 | (l)<br>20.0 | (m)<br>34.7 | (n)<br>1.0            | (o)<br>230 | (p)<br>0.607 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>19.4            | (c)<br>79.6     | (d)<br>65.2 | (e)<br>76.2 | (f)<br>63.6 | (g)<br>73.0 | (h)<br>61.8 | (i)<br>67.4         | (j)<br>75.0 | (k)<br>65.4 | (l)<br>72.8 | (m)<br>63.7 | (n)<br>71.0 | (o)<br>7.8           | (p)<br>140 |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |             |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%          |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP          | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>65.0                     | (b)<br>93.0 | (c)<br>70.4 | (d)<br>62.7 | (e)<br>85.6 | (f)<br>68.5 | (g)<br>60.7 | (h)<br>79.8 | (i)<br>66.9 | (j)<br>31.9   | (k)<br>75.5 | (l)<br>30.4 | (m)<br>72.5 | (n)<br>29.1 | (o)<br>71.5 | (p)<br>73.9       |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 18.3              | 16.1 | 14.3 | -9.4                       | 83.4 | 8.3                | 3.2 | -15.4                                              | 85.7 | -20.3      | 87.6 | -24.9      | 89.4 | -31.0      | 91.8 |  |  |
|                   |      |      | -11.8                      | 69.7 | 6.5                | 2.8 | -16.5                                              | 71.7 | -20.3      | 73.4 | -23.9      | 74.9 | -28.6      | 76.9 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 43.0   | 26.5  | 26.8  | 31.2  | 39.9  | 49.3  | 56.4  | 62.2  | 59.9  | 52.6  | 42.8  | 36.3  | 30.7 |  |  |
|                                                                        | DBStd       | 14.30  | 9.93  | 8.97  | 8.77  | 5.99  | 6.36  | 5.30  | 4.78  | 4.77  | 5.58  | 6.58  | 6.73  | 9.00 |  |  |
|                                                                        | HDD50       | 3654   | 729   | 650   | 583   | 307   | 92    | 9     | 0     | 1     | 37    | 235   | 412   | 599  |  |  |
|                                                                        | HDD65       | 8083   | 1194  | 1070  | 1048  | 753   | 488   | 261   | 116   | 169   | 371   | 687   | 862   | 1064 |  |  |
|                                                                        | CDD50       | 1089   | 0     | 0     | 0     | 3     | 69    | 201   | 377   | 309   | 117   | 13    | 0     | 0    |  |  |
|                                                                        | CDD65       | 44     | 0     | 0     | 0     | 0     | 0     | 3     | 29    | 12    | 0     | 0     | 0     | 0    |  |  |
|                                                                        | CDH74       | 536    | 0     | 0     | 0     | 0     | 12    | 64    | 360   | 99    | 1     | 0     | 0     | 0    |  |  |
|                                                                        | CDH80       | 77     | 0     | 0     | 0     | 0     | 0     | 2     | 64    | 11    | 0     | 0     | 0     | 0    |  |  |
|                                                                        | WSAvg       | 6.9    | 6.9   | 6.8   | 7.1   | 7.0   | 7.3   | 7.2   | 6.6   | 6.1   | 6.8   | 6.9   | 7.1   | 7.3  |  |  |
|                                                                        | PrecAvg     | 24.0   | 1.7   | 1.4   | 1.2   | 1.4   | 1.5   | 2.3   | 2.6   | 2.8   | 2.3   | 2.2   | 2.3   | 2.1  |  |  |
|                                                                        | PrecMax     | 30.6   | 3.3   | 3.0   | 3.0   | 3.5   | 3.9   | 4.4   | 5.0   | 7.0   | 6.9   | 4.7   | 5.3   | 5.0  |  |  |
| (15) Precipitation                                                     | PrecMin     | 16.9   | 0.3   | 0.3   | 0.3   | 0.2   | 0.5   | 0.7   | 0.7   | 0.6   | 0.3   | 0.3   | 0.4   | 0.4  |  |  |
|                                                                        | PrecStd     | 2.7    | 0.8   | 0.5   | 0.6   | 0.9   | 0.7   | 1.1   | 1.3   | 1.6   | 1.5   | 1.0   | 1.1   | 1.0  |  |  |
|                                                                        | DB          | 45.5   | 46.3  | 56.6  | 65.2  | 75.6  | 79.7  | 85.5  | 81.0  | 71.8  | 60.3  | 52.1  | 50.7  |      |  |  |
|                                                                        | MCWB        | 42.5   | 41.4  | 46.2  | 50.9  | 59.6  | 63.5  | 66.1  | 67.5  | 62.0  | 54.9  | 49.1  | 47.6  |      |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 42.3   | 42.8  | 50.7  | 61.2  | 71.3  | 75.8  | 81.6  | 76.7  | 68.4  | 57.4  | 49.5  | 46.1  |      |  |  |
|                                                                        | 2% MCWB     | 40.0   | 39.3  | 42.5  | 48.9  | 57.2  | 61.8  | 65.8  | 65.8  | 59.7  | 53.7  | 47.5  | 43.6  |      |  |  |
|                                                                        | 5% DB       | 40.3   | 40.2  | 46.9  | 57.1  | 67.6  | 72.4  | 78.1  | 73.2  | 65.5  | 55.0  | 47.5  | 43.6  |      |  |  |
|                                                                        | 5% MCWB     | 38.4   | 37.8  | 40.8  | 47.3  | 55.8  | 59.3  | 64.7  | 63.7  | 58.4  | 52.1  | 45.8  | 41.5  |      |  |  |
|                                                                        | 10% DB      | 38.2   | 38.1  | 43.0  | 52.9  | 63.2  | 69.3  | 74.6  | 70.5  | 62.9  | 53.1  | 45.8  | 41.5  |      |  |  |
|                                                                        | 10% MCWB    | 36.7   | 36.3  | 38.2  | 44.2  | 52.9  | 57.9  | 63.1  | 62.1  | 56.9  | 50.5  | 44.2  | 39.8  |      |  |  |
|                                                                        | 0.4% WB     | 43.1   | 43.4  | 46.6  | 52.9  | 61.9  | 66.1  | 70.8  | 71.3  | 63.2  | 56.4  | 49.4  | 48.0  |      |  |  |
|                                                                        | 0.4% MCDB   | 44.4   | 45.6  | 54.0  | 62.6  | 72.1  | 75.3  | 78.1  | 78.1  | 68.6  | 58.6  | 50.6  | 50.2  |      |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 40.5   | 39.9  | 43.4  | 50.1  | 59.1  | 63.1  | 68.0  | 67.2  | 61.1  | 54.2  | 47.6  | 44.2  |      |  |  |
|                                                                        | 2% MCDB     | 41.9   | 41.8  | 49.2  | 59.2  | 67.8  | 72.7  | 76.7  | 73.0  | 66.4  | 56.4  | 48.9  | 45.6  |      |  |  |
|                                                                        | 5% WB       | 38.6   | 38.2  | 41.1  | 47.6  | 56.4  | 61.0  | 66.1  | 65.2  | 59.5  | 52.6  | 45.8  | 41.9  |      |  |  |
|                                                                        | 5% MCDB     | 40.0   | 39.9  | 46.1  | 55.6  | 65.3  | 69.8  | 74.3  | 70.9  | 64.1  | 54.6  | 47.1  | 43.2  |      |  |  |
|                                                                        | 10% WB      | 36.7   | 36.3  | 39.0  | 45.2  | 54.1  | 59.1  | 64.4  | 63.5  | 57.7  | 50.8  | 44.3  | 40.1  |      |  |  |
|                                                                        | 10% MCDB    | 38.1   | 38.0  | 42.4  | 51.8  | 61.9  | 66.8  | 73.2  | 68.9  | 61.8  | 52.6  | 45.5  | 41.5  |      |  |  |
|                                                                        | MDBR        | 10.5   | 12.2  | 15.8  | 20.1  | 20.5  | 19.5  | 19.4  | 18.8  | 17.1  | 13.2  | 9.1   | 9.4   |      |  |  |
|                                                                        | 5% DB       | 9.4    | 12.2  | 20.7  | 28.2  | 27.5  | 26.1  | 26.0  | 22.8  | 21.1  | 14.4  | 10.8  | 9.1   |      |  |  |
| (36) Mean Daily Temperature Range                                      | MCWBR       | 9.0    | 9.9   | 13.9  | 17.3  | 15.7  | 13.5  | 12.0  | 12.9  | 13.5  | 11.2  | 10.1  | 8.7   |      |  |  |
|                                                                        | MCDBR       | 9.0    | 10.7  | 19.0  | 25.6  | 23.8  | 22.7  | 21.1  | 19.6  | 19.0  | 12.6  | 10.2  | 8.9   |      |  |  |
|                                                                        | MCWBR       | 8.8    | 9.0   | 13.3  | 16.3  | 14.5  | 12.7  | 10.9  | 11.8  | 12.7  | 10.4  | 9.7   | 8.7   |      |  |  |
|                                                                        | MCDBR       | 8.8    | 9.0   | 13.3  | 16.3  | 14.5  | 12.7  | 10.9  | 11.8  | 12.7  | 10.4  | 9.7   | 8.7   |      |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.284  | 0.321 | 0.339 | 0.376 | 0.372 | 0.362 | 0.390 | 0.381 | 0.348 | 0.337 | 0.310 | 0.269 |      |  |  |
|                                                                        | taud        | 2.227  | 2.292 | 2.358 | 2.325 | 2.401 | 2.461 | 2.404 | 2.423 | 2.498 | 2.419 | 2.260 | 2.148 |      |  |  |
|                                                                        | Ebn at Noon | 166    | 219   | 253   | 262   | 272   | 277   | 265   | 259   | 250   | 213   | 152   | 124   |      |  |  |
|                                                                        | Edh at Noon | 14     | 22    | 28    | 34    | 33    | 32    | 33    | 30    | 24    | 19    | 14    | 10    |      |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 96     | 282   | 763   | 1273  | 1687  | 1871  | 1725  | 1365  | 889   | 391   | 126   | 65    |      |  |  |
|                                                                        | RadStd      | 14     | 46    | 84    | 121   | 141   | 103   | 161   | 122   | 98    | 62    | 28    | 8     |      |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (52 neighbors) |  | +0.65 | N/A     | N/A    | N/A     | N/A   | N/A   | -209        | -261  | N/A   | N/A   |

Nomenclature: See separate page